OS ORANGE ASSIGNMENT

NAME: Ashwath Rohan
SRN: PES2UG24CS091
Github link : https://github.com/Asro091/PES2UG24CS091-pes-vcs

Screenshots:

Phase 1

```
vboxuser@ashwath:/media/sf_Jackfruit/Orange-2/os-u4-orange-problem$ ./test_objects
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
```

```
vboxuser@ashwath:/media/sf_Jackfruit/Orange-2/os-u4-orange-problem$ find .pes/objects -type f
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
```

Phase 2

```
vboxuser@ashwath:/media/sf_Jackfruit/Orange-2/os-u4-orange-problem$ ./test_tree
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
vboxuser@ashwath:/media/sf_Jackfruit/Orange-2/os-u4-orange-problem$
```

```
vboxuser@ashwath:/media/sf_Jackfruit/Orange-2/os-u4-orange-problem$ xxd "$TREE_FILE" | head -20
00000000: 626c 6f62 2035 0074 6573 740a            blob 5.test.
vboxuser@ashwath:/media/sf_Jackfruit/Orange-2/os-u4-orange-problem$
```

Phase 3

```
vboxuser@ashwath:/media/sf_Jackfruit/Orange-2/os-u4-orange-problem$ make test-integration
=== Running integration tests ===
bash test_sequence.sh
=== PES-VCS Integration Test ===

--- Repository Initialization ---
Initialized empty PES repository in .pes/
PASS: .pes/objects exists
PASS: .pes/refs/heads exists
PASS: .pes/HEAD exists

--- Staging Files ---
Status after add:
Staged changes:
  staged:     file.txt
  staged:     hello.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  (nothing to show)


--- First Commit ---
Committed: ab8fcc834471... Initial commit

Log after first commit:
commit ab8fcc834471b10b43a4b48327725fe648c8b78c46c131084cb13b2f522dd8a1
Author: PES User <pes@localhost>
Date:   1776445289

    Initial commit


--- Second Commit ---
Committed: 3ac640828b58... Update file.txt
```

```
--- Third Commit ---
Committed: 55bd9e208a00... Add farewell

--- Full History ---
commit 55bd9e208a00466283f07b81d85e12ddd1a73fc744b253ea6763609b266cb435
Author: PES User <pes@localhost>
Date:   1776445289

    Add farewell

commit 3ac640828b58623b36ed3b3a37b7462481fe416d48e7635b2569b1c2e297f2cb
Author: PES User <pes@localhost>
Date:   1776445289

    Update file.txt

commit ab8fcc834471b10b43a4b48327725fe648c8b78c46c131084cb13b2f522dd8a1
Author: PES User <pes@localhost>
Date:   1776445289

    Initial committial commiQ


--- Reference Chain ---
HEAD:
ref: refs/heads/main
refs/heads/main:
55bd9e208a00466283f07b81d85e12ddd1a73fc744b253ea6763609b266cb435

--- Object Store ---
Objects created:
10
.pes/objects/0b/d69098bd9b9cc5934a610ab65da429b525361147faa7b5b922919e9a23143d
.pes/objects/10/1f28a12274233d119fd3c8d7c7216054ddb5605f3bae21c6fb6ee3c4c7cbfa
.pes/objects/3a/c640828b58623b36ed3b3a37b7462481fe416d48e7635b2569b1c2e297f2cb
.pes/objects/55/bd9e208a00466283f07b81d85e12ddd1a73fc744b253ea6763609b266cb435
.pes/objects/58/a67ed1c161a4e89a110968310fe31e39920ef68d4c7c7e0d669579753 3f50d
```

```
--- Reference Chain ---
HEAD:
ref: refs/heads/main
refs/heads/main:
55bd9e208a00466283f07b81d85e12ddd1a73fc744b253ea6763609b266cb435

--- Object Store ---
Objects created:
10
.pes/objects/0b/d69098bd9b9cc5934a610ab65da429b525361147faa7b5b922919e9a23143d
.pes/objects/10/1f28a12274233d119fd3c8d7c7216054ddb5605f3bae21c6fb6ee3c4c7cbfa
.pes/objects/3a/c640828b58623b36ed3b3a37b7462481fe416d48e7635b2569b1c2e297f2cb
.pes/objects/55/bd9e208a00466283f07b81d85e12ddd1a73fc744b253ea6763609b266cb435
.pes/objects/58/a67ed1c161a4e89a110968310fe31e39920ef68d4c7c7e0d6695797533f50d
.pes/objects/ab/5824a9ec1ef505b5480b3e37cd50d9c80be55012cabe0ca572dbf959788299
.pes/objects/ab/8fcc834471b10b43a4b48327725fe648c8b78c46c131084cb13b2f522dd8a1
.pes/objects/b1/7b838c5951aa88c09635c5895ef7e08f7fa1974d901ce282f30e08de0ccd92
.pes/objects/d0/7733c25d2d137b7574be8c5542b562bf48bafeaa3829f61f75b8d10d5350f9
.pes/objects/db/07a1451ca9544dbf66d769b505377d765efae7adc6b97b75cc9d2b3b3da6ff

=== All integration tests completed ===
vboxuser@ashwath:/media/sf_Jackfruit/Orange-2/os-u4-orange-problem$
```

Phase 4

```
vboxuser@ashwath:/media/sf_Jackfruit/Orange-2/os-u4-orange-problem$ export PES_AUTHOR="Ashwath Rohan <PES2UG24CS091>"
vboxuser@ashwath:/media/sf_Jackfruit/Orange-2/os-u4-orange-problem$ ./pes init
Initialized empty PES repository in .pes/
vboxuser@ashwath:/media/sf_Jackfruit/Orange-2/os-u4-orange-problem$ echo "Hello, World!" > hello.txt
echo "# My Project" > README.md
vboxuser@ashwath:/media/sf_Jackfruit/Orange-2/os-u4-orange-problem$ ./pes add hello.txt README.md
vboxuser@ashwath:/media/sf_Jackfruit/Orange-2/os-u4-orange-problem$ ./pes status
Staged changes:
  staged:     README.md
  staged:     hello.txt
  staged:     test.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  untracked:  .gitignore
  untracked:  commit.c
  untracked:  commit.h
  untracked:  index.c
  untracked:  index.h
  untracked:  Makefile
  untracked:  object.c
  untracked:  pes.c
  untracked:  pes.h
  untracked:  test_objects.c
  untracked:  test_sequence.sh
  untracked:  test_tree.c
  untracked:  tree.c
  untracked:  tree.h

vboxuser@ashwath:/media/sf_Jackfruit/Orange-2/os-u4-orange-problem$
```

```
vboxuser@ashwath:/media/sf_Jackfruit/Orange-2/os-u4-orange-problem$ export PES_AUTHOR="Ashwath Rohan <PES2UG24CS091>"
vboxuser@ashwath:/media/sf_Jackfruit/Orange-2/os-u4-orange-problem$ ./pes init
Initialized empty PES repository in .pes/
vboxuser@ashwath:/media/sf_Jackfruit/Orange-2/os-u4-orange-problem$ echo "Hello, World!" > hello.txt
echo "# My Project" > README.md
vboxuser@ashwath:/media/sf_Jackfruit/Orange-2/os-u4-orange-problem$ ./pes add hello.txt README.md
vboxuser@ashwath:/media/sf_Jackfruit/Orange-2/os-u4-orange-problem$ ./pes status
Staged changes:
  staged:     README.md
  staged:     hello.txt
  staged:     test.txt

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Untracked files:
  untracked:  .gitignore
  untracked:  commit.c
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  untracked:  index.c
  untracked:  index.h
  untracked:  Makefile
  untracked:  object.c
  untracked:  pes.c
  untracked:  pes.h
  untracked:  test_objects.c
  untracked:  test_sequence.sh
  untracked:  test_tree.c
  untracked:  tree.c
  untracked:  tree.h

vboxuser@ashwath:/media/sf_Jackfruit/Orange-2/os-u4-orange-problem$ ./pes commit -m "Initial commit"
Committed: fc752aa1aa5e... Initial commit
vboxuser@ashwath:/media/sf_Jackfruit/Orange-2/os-u4-orange-problem$
```

```
vboxuser@ashwath:/media/sf_Jackfruit/Orange-2/os-u4-orange-problem$ ./pes log
commit fc752aa1aa5e7ff60bdb5b75afa579e92f7e0f01e1d905c8b03324cc7a928e8d
Author: Ashwath Rohan <PES2UG24CS091>
Date:   1776445810

    Initial commit◆

vboxuser@ashwath:/media/sf_Jackfruit/Orange-2/os-u4-orange-problem$
```

```
vboxuser@ashwath:/media/sf_Jackfruit/Orange-2/os-u4-orange-problem$ ./pes log
commit fc752aa1aa5e7ff60bdb5b75afa579e92f7e0f01e1d905c8b03324cc7a928e8d
Author: Ashwath Rohan <PES2UG24CS091>
Date:   1776445810

    Initial commit◆

vboxuser@ashwath:/media/sf_Jackfruit/Orange-2/os-u4-orange-problem$ echo "More content" >> hello.txt
vboxuser@ashwath:/media/sf_Jackfruit/Orange-2/os-u4-orange-problem$ ./pes add hello.txt
./pes commit -m "Update hello.txt"
Committed: e08b704dd66d... Update hello.txt
vboxuser@ashwath:/media/sf_Jackfruit/Orange-2/os-u4-orange-problem$ echo "Goodbye!" > bye.txt
./pes add bye.txt
./pes commit -m "Add farewell"
Committed: 8cdb50dea98c... Add farewell
vboxuser@ashwath:/media/sf_Jackfruit/Orange-2/os-u4-orange-problem$ ./pes log
commit 8cdb50dea98ccbd9a392b5020569959658fe91862df646a17a6083602be9b55a
Author: Ashwath Rohan <PES2UG24CS091>
Date:   1776446098

    Add farewell

commit e08b704dd66dc94c5e64cd174cd27c537a609aaadbe11b065211548af6f91e88
Author: Ashwath Rohan <PES2UG24CS091>
Date:   1776446088

    Update hello.txtinsta !

commit fc752aa1aa5e7ff60bdb5b75afa579e92f7e0f01e1d905c8b03324cc7a928e8d
Author: Ashwath Rohan <PES2UG24CS091>
Date:   1776445810

    Initial commitQ

vboxuser@ashwath:/media/sf_Jackfruit/Orange-2/os-u4-orange-problem$
```

```
vboxuser@ashwath:/media/sf_Jackfruit/Orange-2/os-u4-orange-problem$ export PES_AUTHOR="Ashwath Rohan <PES2UG24CS091>"
vboxuser@ashwath:/media/sf_Jackfruit/Orange-2/os-u4-orange-problem$ ./pes init
Initialized empty PES repository in .pes/
vboxuser@ashwath:/media/sf_Jackfruit/Orange-2/os-u4-orange-problem$ echo "version 1.0" > version.txt
echo "print('Hello')" > main.py
./pes add version.txt main.py
vboxuser@ashwath:/media/sf_Jackfruit/Orange-2/os-u4-orange-problem$ ./pes commit -m "Initial project setup"
Committed: f88b5576b6c5... Initial project setup
vboxuser@ashwath:/media/sf_Jackfruit/Orange-2/os-u4-orange-problem$ echo "version 1.1" > version.txt
./pes add version.txt
./pes commit -m "Bump version"
Committed: 07fac949be5d... Bump version
vboxuser@ashwath:/media/sf_Jackfruit/Orange-2/os-u4-orange-problem$ ./pes log
commit 07fac949be5d8a5f1a81cc7345cd8a82250eb843c570862c68092e668edf1548
Author: Ashwath Rohan <PES2UG24CS091>
Date:   1776446234

    Bump version

commit f88b5576b6c51b43c75e48d20f85d68dc008380429ec77b3e579b2d73401536e
Author: Ashwath Rohan <PES2UG24CS091>
Date:   1776446228

    Initial project setup !

commit 8cdb50dea98ccbd9a392b5020569959658fe91862df646a17a6083602be9b55a
Author: Ashwath Rohan <PES2UG24CS091>
Date:   1776446098

    Add farewellect setup !

commit e08b704dd66dc94c5e64cd174cd27c537a609aaadbe11b065211548af6f91e88
Author: Ashwath Rohan <PES2UG24CS091>
Date:   1776446088

    Update hello.txtsetup !

commit fc752aa1aa5e7ff60bdb5b75afa579e92f7e0f01e1d905c8b03324cc7a928e8d
Author: Ashwath Rohan <PES2UG24CS091>
```

```
commit fc752aa1aa5e7ff60bdb5b75afa579e92f7e0f01e1d905c8b03324cc7a928e8d
Author: Ashwath Rohan <PES2UG24CS091>
Date:   1776445810

    Initial commitQ
```

```
vboxuser@ashwath:/media/sf_Jackfruit/Orange-2/os-u4-orange-problem$ echo "=== Repository Structure ==="
find .pes -type f | sort

echo "=== HEAD Reference ==="
cat .pes/HEAD

echo "=== Main Branch ==="
cat .pes/refs/heads/main

echo "=== Index Contents ==="
cat .pes/index
=== Repository Structure ===
.pes/HEAD
.pes/index
.pes/objects/07/fac949be5d8a5f1a81cc7345cd8a82250eb843c570862c68092e668edf1548
.pes/objects/27/6b3188aa18228a60323b0bed07255555f4ed076d0b230cb27df8c79fef4f28
.pes/objects/3d/8b77d81ea4c9ad30ced446b9ae9a2f3e4ff661aa61efe1e7ef0a61b34df36b
.pes/objects/5c/1c11aac7a51b3bbf39d245c25fa50822b31107f7e3f592f417cdf826b046b8
.pes/objects/64/5a9d5586460c63d7242369cecb0c8a8b461453abca4bb37284e5bddda9d0d1
.pes/objects/81/7403b7ccbaa0091ef916f46bddd67ca8dfe83c6fe804cafc4114c9008afbf8
.pes/objects/82/11a5480614a8330b294a3b4c8fd31f62b39298776c4d8434b7ebbdb7613e9b
.pes/objects/85/c57eb8f2b83a3858249b5f7501d7e90772287e927bc9ce4a57d113ebd21256
.pes/objects/8c/db50dea98ccbd9a392b5020569959658fe91862df646a17a6083602be9b55a
.pes/objects/8d/a45a498de12b2badd0fa02cafc4fc6264dc182fbcede89741381bf34f6b6e1
.pes/objects/99/9f24152159e51756a944d32257bf22080ff8608fff87ca9a4a823764e13dbe
.pes/objects/c8/b19d874517a11bf1463a0a54a3344d7e60982d341cc8ae6b080a3627f1c3db
.pes/objects/da/bc789f60c22621c92df8736ff8cb60e35185584772b93b9315a3e2aab55653
.pes/objects/e0/8b704dd66dc94c5e64cd174cd27c537a609aaadbe11b065211548af6f91e88
.pes/objects/e5/298b2642a616d55c74298ee206da3ef2be79c334592e0fc4bd5328f08bbcf0
.pes/objects/eb/04b994c55decc2a0d7ceee6b2b8b18d6277bd1ed0f4451e6613261209bddb9
.pes/objects/f8/8b5576b6c51b43c75e48d20f85d68dc008380429ec77b3e579b2d73401536e
.pes/objects/fc/752aa1aa5e7ff60bdb5b75afa579e92f7e0f01e1d905c8b03324cc7a928e8d
.pes/refs/heads/main
=== HEAD Reference ===
ref: refs/heads/main
=== Main Branch ===
07fac949be5d8a5f1a81cc7345cd8a82250eb843c570862c68092e668edf1548
```

```
=== Main Branch ===
07fac949be5d8a5f1a81cc7345cd8a82250eb843c570862c68092e668edf1548
=== Index Contents ===
100755 8da45a498de12b2badd0fa02cafc4fc6264dc182fbcede89741381bf34f6b6e1 1776445627 13 README.md
100755 276b3188aa18228a60323b0bed07255555f4ed076d0b230cb27df8c79fef4f28 1776446098 9 bye.txt
100755 5c1c11aac7a51b3bbf39d245c25fa50822b31107f7e3f592f417cdf826b046b8 1776446080 27 hello.txt
100755 8211a5480614a8330b294a3b4c8fd31f62b39298776c4d8434b7ebbdb7613e9b 1776446213 15 main.py
100755 999f24152159e51756a944d32257bf22080ff8608fff87ca9a4a823764e13dbe 1776445077 5 test.txt
100755 645a9d5586460c63d7242369cecb0c8a8b461453abca4bb37284e5bddda9d0d1 1776446234 12 version.txt
vboxuser@ashwath:/media/sf_Jackfruit/Orange-2/os-u4-orange-problem$
```

Phase 5 and 6

Analysis question answers

Q 5.1

Update .pes pointers by verifying the branch file exists at .pes/refs/heads/, reading the target commit hash, and rewriting .pes/HEAD to ref: refs/heads/ using an atomic temp + rename. Resolve the target snapshot by reading the commit object, its root tree, and recursively

expanding tree objects to build the full path → blob mapping.

Update the working directory by writing each blob to its file with correct mode, removing tracked files not present in the target tree, and recreating directories as needed. Rebuild the index by setting .pes/index entries to match the target tree with updated stat metadata.

This is complex because it must safely handle deletions, nested directories, permissions, partial failures, and avoid data loss when uncommitted changes overlap with target changes.

Q 5.2

Use a 3-way comparison per tracked file with base as the current HEAD tree blob hash, staged intent as the hash in .pes/index, and working file as stat plus recomputed blob hash. Build the target tree mapping and for each path differing between current HEAD and target, check if the working directory is dirty relative to the current committed state. If the file is dirty and the target also changes that path, refuse checkout. Also detect untracked file overwrite cases where an untracked working file would be overwritten by the target tree and refuse. This enforces the rule that checkout should not overwrite local non-committed content.

Q 5.3

Detached HEAD means HEAD points directly to a commit hash rather than to a branch reference. In this state, new commits can still be created, but no branch moves forward automatically. These commits remain valid objects in the repository, but they may become unreachable if no branch references them. Recovery is possible by creating a new branch pointing to the detached commit before garbage collection removes it.

Q 6.1

Use a mark-and-sweep approach where the mark phase initializes a hash set, seeds it with all branch tips, and performs DFS or BFS traversal marking commits, trees, and blobs while enqueueing referenced objects. The sweep phase scans all files in .pes/objects and deletes those whose hashes are not in the reachable set, optionally cleaning empty directories. A hash set ensures O(1) expected membership checks and deduplication during traversal. In scale, traversal visits unique reachable objects, typically hundreds of thousands to a few million, making the complexity proportional to reachable objects plus total objects on disk.

Q 6.2

A race occurs when commit writes blob and tree objects, then commit object, and finally updates the branch ref while GC runs before the ref update and cannot see the new chain. GC treats new objects as unreachable and deletes them, after which the branch points to a broken history with missing objects. Real systems avoid this through synchronization and locking around ref updates and GC, reachability protections like reflogs and grace periods, avoiding immediate pruning of recent objects, and ensuring objects are durable before references move.